

Simulating nitrate leaching under winter wheat grown on a structured clay soil considering bypass flow

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Abstract

Nitrate leaching from drained plots of structured clay soil under winter wheat is simulated for one growing season using the SOILN model. Results are compared with field measurements from two replicate plots. Soil water movement is simulated both with and without a bypass flow component using the SOIL model. Flow to field drains and soil water content in the root zone are simulated better when bypass flow is included. The results emphasise the importance of considering bypass flow in modelling leaching from structured clay soils. Simulations of nitrate leaching using the SOILN model show that the model captures the main features of the cumulative loss of nitrate over the year.

1. Introduction

Nitrate leaching from agriculture is a major contributor to high nitrate concentrations in some surface and groundwaters in the United Kingdom. This is of amenity and public health concern, and the subject of EU legislation. Nitrate is highly mobile and if present in the soil will readily drain in solution. Thus freely draining soils tend to be more susceptible to leaching than less permeable soils. However, where low permeability soils are drained and liable to cracking, there may be circumstances in which there is rapid movement of water through the drainage system from near-surface soil layers with a high nitrate content. Understanding the influence on nitrate leaching of drainage under these conditions is important since such soils form significant areas of productive agricultural land in the United Kingdom.

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Many of the major processes occurring in macropore flow to field drains are now included in models of soil water movement and the transport of solutes. If such models can be shown to simulate effectively the leaching of nitrate in field experiments, they may be further used for prediction at different locations and under different agricultural managements. One such model is SOILN (Johnsson et al., 1987), used to simulate field-scale nitrate leaching in conjunction with the SOIL model (Jansson, 1991) which simulates soil water movement. These models have been used to simulate nitrate leaching at Brimstone Farm, an experimental site in southern England. Simulated results have been compared with field measurements of leaching for two replicate plots.

The objectives of this study were to quantify bypass flow and to investigate its significance in improving the prediction of the soil water balance components and nitrate leaching to the drainage system.

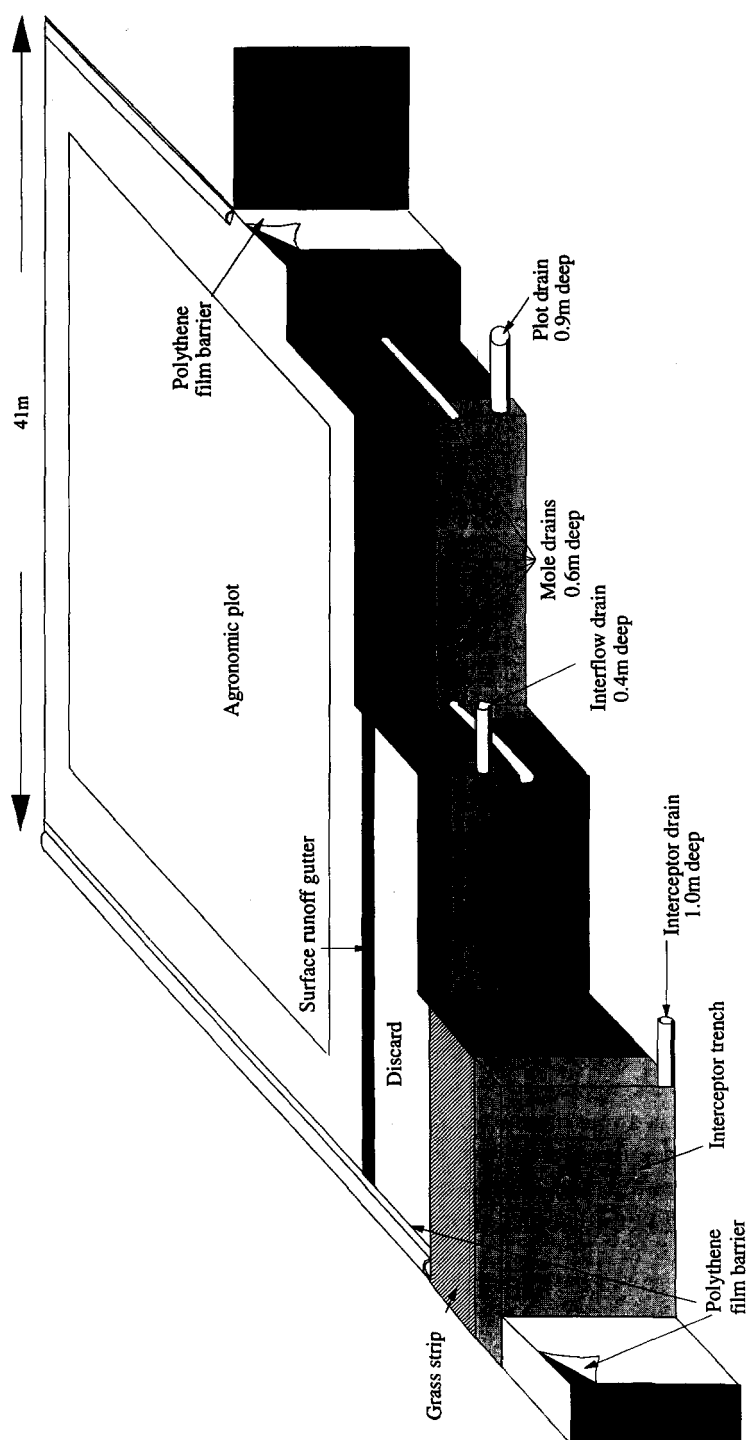
2. Experimental site

The experimental site at Brimstone Farm, Faringdon, Oxfordshire was established by Agriculture and Food Research Council (AFRC) and Agricultural Development Advisory Service (ADAS) in 1978 on a heavy clay soil of the Denchworth series. The initial aim of research carried out there was to investigate the effects of different field drainage systems on crop production and water losses in a heavy soil typical of arable land in the UK. Direct drilled and ploughed land were compared, with water and nitrate losses from the field drainage system measured.

The field site at Brimstone Farm comprises 20 isolated plots large enough for normal agricultural operations. Each plot measures 59 m by 41 m, with a central agronomic area of 40 m by 28 m. The plot experimental areas are hydrologically isolated to a depth of 1.1 m down the slope using a continuous polythene membrane placed in an open trench in the discard area between plots. Water moving down slope is intercepted by pipe drain trenches, 0.1 m wide, located at 1.0 m depth in the discard area on the lower side of each plot and filled with permeable backfill material to the field surface. The hydrological area that is monitored covers the full plot width of 41 m and extends 46 m downslope (2% slope) from the interceptor trench at the top of each plot to the plot drain (Fig. 1). The remaining 13 m of the plot has no subsurface drainage.

The analysis described here relates to field measurements of nitrate leaching carried out during the 1985–86 crop year, for which data suitable for use with SOILN were available. The modelling is retrospective, so could only make use of historic data which were not collected specifically to support modelling.

During 1985 the area had been laid out as five replicates of two tillage treatments by two drainage treatments, and sown with winter wheat following oil-seed rape. Plots were either direct drilled or ploughed, and mole drainage was either present or absent. Mole drains were at 0.6 m depth, nominally 0.1 m in diameter and drawn at right angles to a plot drain installed at 0.9 m depth at the lower boundary of the plot. There was a total of 19 mole drains 2 m apart for each drained plot. Since the plots were also



ploughed to a depth of 0.2 m an interflow pipe drain to intercept plough layer flow was installed. This was located at a depth of 0.35 m and 3 m downslope of the plot drain, and covered with permeable backfill in a polythene lined trench.

The combination of treatments studied is restricted to mole drainage with ploughing. Data for two of the five replicates are used for model comparisons.

The hydrology of the site has been extensively studied. Harris et al. (1993) reported that the lag time between peak rainfall and peak drainflow for drained plots is typically 1.5–3 h, and that the presence of continuous vertical cracks encourages downward movement, particularly in early autumn and late spring.

Haria et al. (1994) and others have shown that earthworm channels can play a key role in water and solute transport in clay soils. However, at Brimstone Farm only a few deep burrowing earth worms were found and they probably have little influence on the drainage. Instead, Goss et al. (1984) showed that peak drainflow was a result of rapid water movement and concluded that the crack structure is important in transmitting water from the topsoil to the subsoil. Castle (1991) suggested that water movement to the drains is possible through the primary leg slot crack produced during mole draining, which provides an adequate connection between the surface and subsoil.

Other flow mechanisms have been observed at plots elsewhere on the same soil series. Haria et al. (1994) report the presence of a perched saturated layer in the A horizon at another site in Southern England, leading to rapid macropore flow to the drains bypassing the matrix of the B horizon. In this case, the drain flow consists of water arriving from above drain depth and not from any rise in the water table. Although there was a perched water table over a clay pan when the experimental plots at Brimstone Farm were created, this was removed successfully by shallow subsoiling (Castle, 1991). Consequently, the subsurface interflows collected from the A horizon over many years were very small (Harris and Howse, 1991). These authors also reported that the water table remained close to the surface, with the profile saturated for longer periods. These reported observations show that care should be exercised in making general inferences for sites even with the same soil series.

3. The SOIL and SOILN models

The SOIL model was chosen to simulate the soil water balance and heat flow at Brimstone Farm. It is a one-dimensional model of vertical soil water movement and heat flow for layered soils covered with vegetation, and operates at a daily time scale. Process descriptions are based on equations expressed in continuous space and time. These are solved by finite difference approximation by regarding the soil as discretised into several layers.

Water flow in the soil in the absence of bypass flow (q_w) is simulated using Darcy's law:

$$q_w = k_w \left[\frac{\partial \psi}{\partial z} + 1 \right] \quad (1)$$

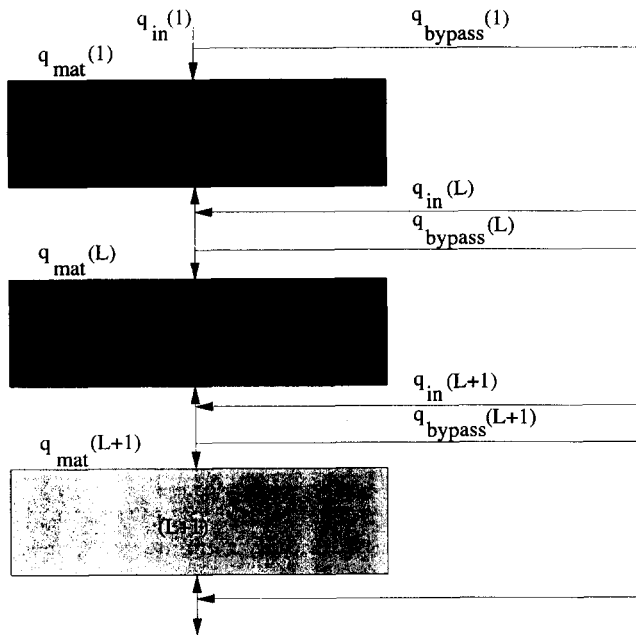


Fig. 2. SOIL model showing bypass flow in macropores (after Jansson, 1991).

where $\psi = \psi(\theta, z)$ is the water tension and $k_w = k_w(\theta, z)$ is the hydraulic conductivity. Eq. (1) is combined with the continuity equation:

$$\frac{\partial \theta}{\partial t} = \frac{\partial q_w}{\partial z} + S_w \quad (2)$$

to give a full description of soil water dynamics. Here S_w is a source/sink term whose main component is plant water uptake and θ is soil water content.

Bypass flow in macropores (Fig. 2) when smaller pores are partially filled with water is simulated separately. The infiltration rate at the soil surface or the vertical flow in the macropores at any depth in the soil profile, q_{in} , determines the partitioning into Darcy flow, q_{mat} , and bypass flow, q_{bypass} :

$$q_{mat} = \max \left(k_w \left[\frac{\partial \psi}{\partial z} + 1 \right], q_{in} \right) \quad 0 < q_{in} < S_{mat} \quad (3)$$

$$q_{bypass} = 0 \quad 0 < q_{in} < S_{mat} \quad (4)$$

$$q_{mat} = S_{mat} \quad q_{in} \geq S_{mat} \quad (5)$$

$$q_{bypass} = q_{in} - q_{mat} \quad q_{in} \geq S_{mat} \quad (6)$$

At the soil surface q_{in} is the influx of water. At other depths in the soil, it is the vertical flow rate in the macropores, q_{bypass} , from layers immediately above. S_{mat} is defined as:

$$S_{mat} = a_{scale} a_r k_{mat} pF \quad (7)$$

where k_{mat} is the maximum conductivity of the matrix pores, a_r is the ratio between compartment thickness and the unit horizontal area represented by the model, pF is $\log_{10}(\psi)$ and a_{scale} is an empirical scaling coefficient accounting for the geometry of the aggregates. The values of a_r and a_{scale} are set to default values suggested as appropriate by Jansson (1991). The calculated water flow in the matrix q_{mat} is used to update the water contents and the water tension while q_{bypass} is directed without delay to the next soil compartment.

Saturated layers are assumed to lose water, notionally to field drains. This loss is taken to be proportional to the hydraulic gradient and to the thickness and saturated hydraulic conductivity of a given soil layer.

The model includes treatment of evapotranspiration. Daily potential transpiration is calculated according to Monteith (1965). Plant water uptake is considered equal to actual transpiration, computed as potential transpiration adjusted for root density, water stress and temperature. Surface evaporation is computed using the Penman combination equation.

The variables computed within SOIL which may be used for model testing include the water content of each soil layer, surface flow and loss of water to field drains. The heat flow component of the model provides daily estimates of soil temperature.

Apart from output test variables, the SOIL model provides the water flow and temperature variables needed to run the SOILN model which simulates transformations between different forms of nitrogen in the soil. These include vertical water flows between layers, infiltration through the soil surface, water loss to drains, surface runoff, soil moisture of each layer and the soil temperature of each layer (Johnsson et al., 1987).

The SOILN model takes as external nitrogen inputs wet and dry atmospheric deposition, inorganic and organic fertiliser applications. Mineral nitrogen derived from the decomposition of plant residues is an additional internal source. Two organic matter pools are considered: a fast cycling pool (litter), representing an organic matter–microbial biomass complex which receives fresh organic material, and a slow cycling pool (humus) composed of stabilised decomposition products. Manure is included as a separate component because it is different in its chemical composition from plant material entering the litter pool.

Incorporation of plant materials into the litter component is simulated twice, at harvest and when the field is ploughed. At ploughing, nitrogen in above-ground harvest residues (and remaining roots in a perennial crop) is incorporated into the plough layer. The carbon incorporated is proportional to the C/N ratio of the above-ground harvest residues (and roots in the case of perennial crops).

Decomposition rates in the two organic pools (litter and manure) are the main controls on N mineralisation from these sources. The net mineralisation is calculated by the balance between the release of nitrogen during decomposition and the nitrogen immobilisation during microbial synthesis and humification. Both ammonium and nitrate can be immobilised but with preference to available ammonium. Decomposition, mineralisation/immobilisation and humus formation in manure are calculated in the same way as for litter. Mineralisation of humus is calculated as a first-order rate.

In simulating nitrification, the mineral N is segregated into nitrate and ammonium by assuming a nitrate/ammonium ratio that is characteristic of a particular soil. The rapid nitrification rate and the small amounts of ammonium often observed in agricultural soils justify this assumption. If the nitrate/ammonium ratio in a certain layer becomes greater than a specified value, no ammonium transformation to nitrate takes place.

Decomposition, mineralisation and nitrification are regulated by the same abiotic response function involving soil temperature and soil moisture. There are optimum temperature and soil moisture values below which the rates are assumed to decrease.

Diffusion and root absorption are assumed to be the main mechanisms behind nitrogen uptake by roots and therefore N-uptake is not associated with water uptake. N-uptake includes both nitrate and ammonium from the root zone.

The denitrification rate of each layer is calculated from the potential denitrification rate, soil water/aeration status (dependent on soil moisture) and the half saturation constant.

The movement of nitrate between soil layers and to the drainage system is in solution and therefore its movement is determined by water flow rates. The nitrate transport through any soil layer is calculated as the product of nitrate concentration and water flow through that layer. Ammonium is assumed to be immobile with regard to leaching.

4. Model application

The SOIL and SOILN models were jointly run at a daily time step from the start of the crop year. Data for model performance testing were available from Plots 7 and 10, each ploughed and mole-drained. Common parameter values, initial conditions and driving variable values were used for each plot.

4.1. Parameters

In applying the SOIL model at the Brimstone Farm site, the soil profile was notionally divided into 12 layers each 10 cm thick. The soil physical characteristics were estimated for each layer. The saturated hydraulic conductivity of the soil matrix in the top 20 cm was taken equal to the steady infiltration rate of 4.2 cm h^{-1} obtained using a double ring infiltrometer. Below that depth the saturated conductivity was obtained from piezometer measurements, as described by Youngs and Goss (1988). Their values decreased from 2.78 at 30 cm to 0.004 cm h^{-1} at 110 cm. The saturated soil moisture content was taken equal to the soil porosity, calculated from the measured bulk density for each layer. The values decreased from 60% in the top 20 cm layer to 45% at 110 cm.

The relationships between soil moisture content, soil water potential and hydraulic conductivity were described by the Brooks and Corey (1964) equations:

$$\theta = \theta_s \left[\frac{\psi_e}{\psi} \right]^b \quad (8)$$

where θ_s is the saturated water content, ψ_e is the air entry potential and b is a soil parameter and:

$$K = K_s \left[\frac{\psi_e}{\psi} \right]^{2+3b} \quad (9)$$

where K_s is the saturated hydraulic conductivity and b is as in Eq. (8).

The soil moisture characteristic curve (Eq. (8)) was computed from in situ data collected in the drying cycle of spring 1988. Laboratory soil measurements gave significantly different soil moisture characteristics and were discarded. The in situ measurements are preferred since they attempt to account for macropore structure which has been found to be important at this site (Catt, 1991). The estimated pore size distribution index, b , was 0.10 for the top 20 cm layer and 0.07 for the lower layers. The water potential at air entry was assigned the value of 40.5 cm suggested by Clapp and Hornberger (1978) for clay soils.

To simulate nitrogen transformations, C/N ratios of 15, 50 and 25 were used for microorganisms and humified products, above ground residues and roots, respectively. The specific mineralisation rates of humus and litter decomposition were set to 0.00005 and 0.05 day⁻¹; the specific nitrification rate to 0.3 day⁻¹; and the potential denitrification rate to 0.4 kg N ha⁻¹ day⁻¹.

There was no calibration of any of the model parameters.

4.2. Initial conditions

The initial soil moisture tension was assumed to be 200 cm of water and soil temperature 12°C throughout the soil profile. Ground water level was assumed to be 1.6 m below the soil surface. This value is taken from measurements made in previous years (Harris et al., 1984).

For each soil layer, the initial NO₃-N load was set to 12.50 kg N ha⁻¹ with an NH₄-N load of 3.0 kg N ha⁻¹ for all layers of the soil profile.

Starting values for initial conditions were selected as likely values, but adjusted by informal calibration.

4.3. Driving variables

Daily measurements of rainfall, air temperature, relative humidity, solar and net radiation and wind speed were used as input to the SOIL model. The rainfall measurements used were made at Brimstone Farm. The air temperature, wind speed, cloud cover, relative humidity and sunshine hours measurements used to estimate evapotranspiration were obtained from the Brize Norton weather station, Oxfordshire, approximately 15 km to the north. Solar and net radiation were calculated from the latitude and sunshine hours.

Crop data needed to estimate actual evaporation, such as leaf area index (LAI) and root depth, were not available. A maximum LAI of 5 was assumed. The maximum root depth was assigned a value of 60 cm below soil surface. The crop growth sub-model uses the heat sums and sowing date to provide continuous estimates of both

LAI and rooting depth up to their specified maximum values. The stomatal resistance was assumed to vary between 20 and 70 s m⁻¹ according to the time of the year (Jansson, 1991).

Winter wheat was sown on October 10 and 130 kg ha⁻¹ ammonium nitrate fertiliser was applied on April 30 1986. Dry and wet deposition estimates were set to 0.006 g N m⁻¹ year⁻¹ and 0.9 mg N l⁻¹, respectively (Goulding, 1990).

5. Results and discussion

5.1. Soil water movement

In the first simulation of the SOIL model, only flow through the soil matrix was considered and bypass flow through macropores was ignored. The simulated and observed drainage from Plot 7 are shown in Fig. 3, which shows that the model did not simulate drainage flow on some occasions, especially during March and April, and it also underestimated the highest peak in December 1985. Results for Plot 10 were similar. As the rainfall and drainflows for Plots 7 and 10 (Figs. 4 and 5) are closely related, with a more rapid response of drainflow than would be expected if water movement were solely through the soil matrix, with its low hydraulic conductivity, some of the water probably reaches the drains through bypass flow, as suggested by Goss et al. (1984) and Castle (1991).

When the SOIL model was re-run to include bypass flow, the simulated drainage was improved, as shown in Fig. 6 for Plot 7 and in Fig. 7 for Plot 10. The high peak in December is better simulated and simulated flows appear during March–April.

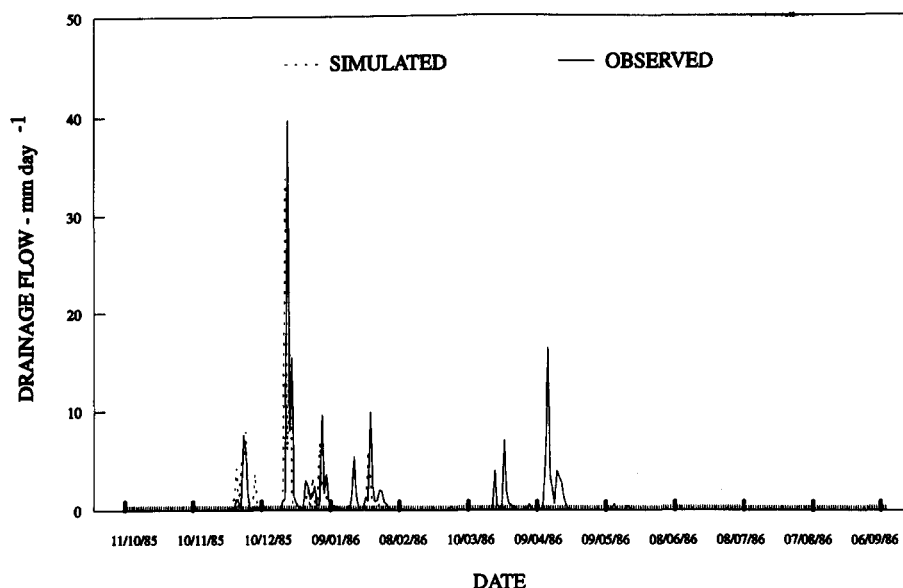


Fig. 3. Measured and simulated drainage flow – Plot 7.

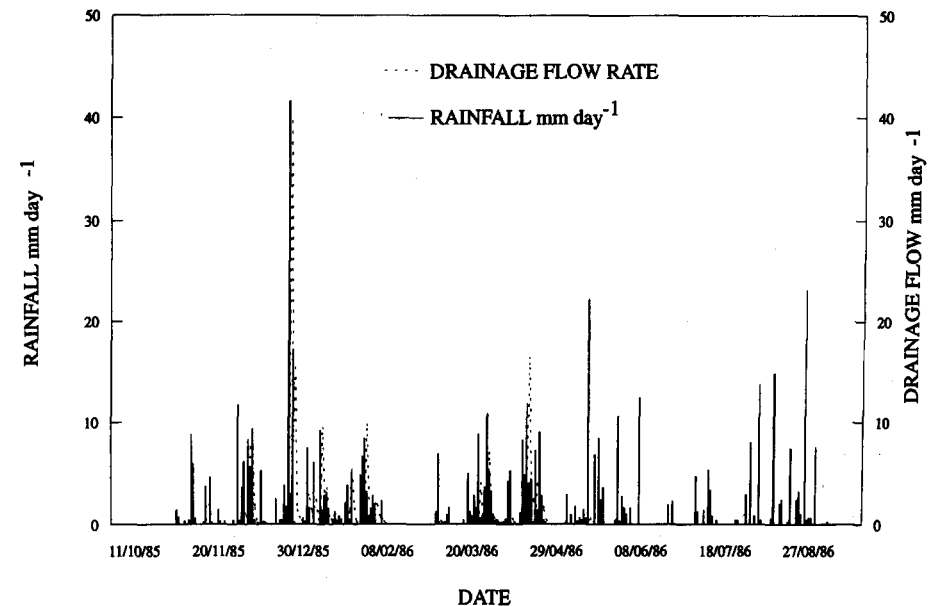


Fig. 4. Measured drainage flow rate and rainfall – Plot 7.

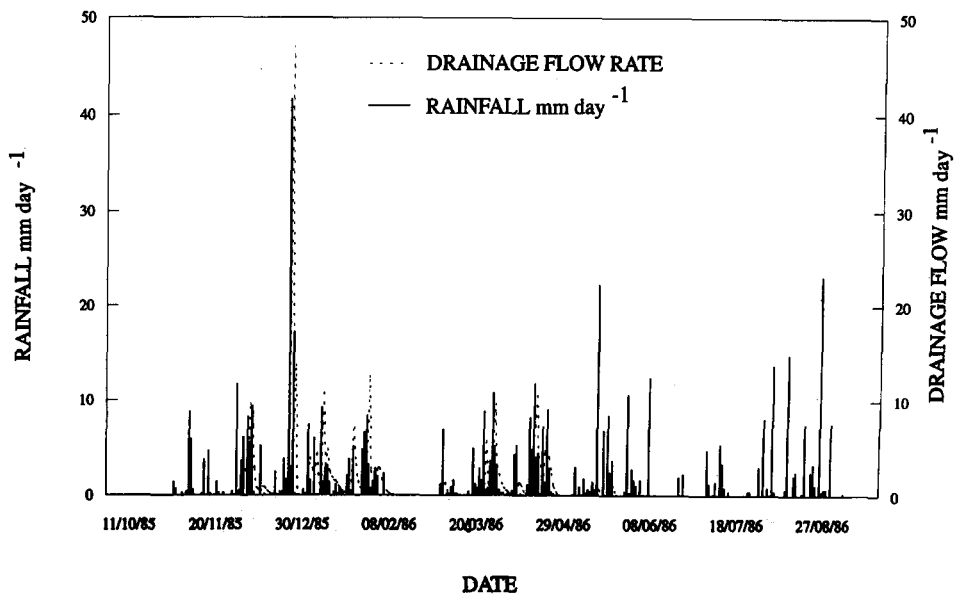


Fig. 5. Measured drainage flow rate and rainfall – Plot 10.

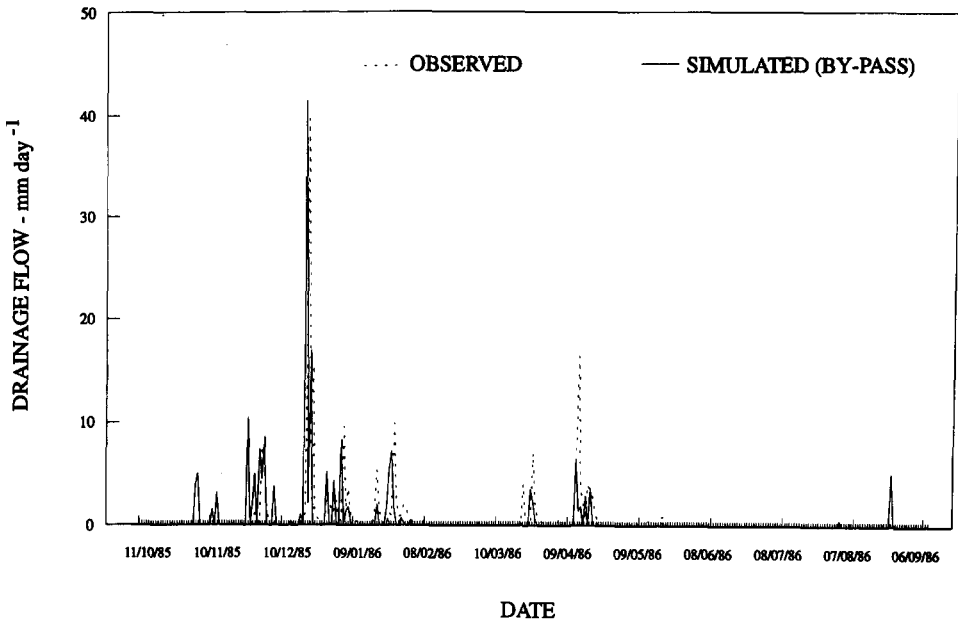


Fig. 6. Measured and simulated drainage with bypass flow – Plot 7.

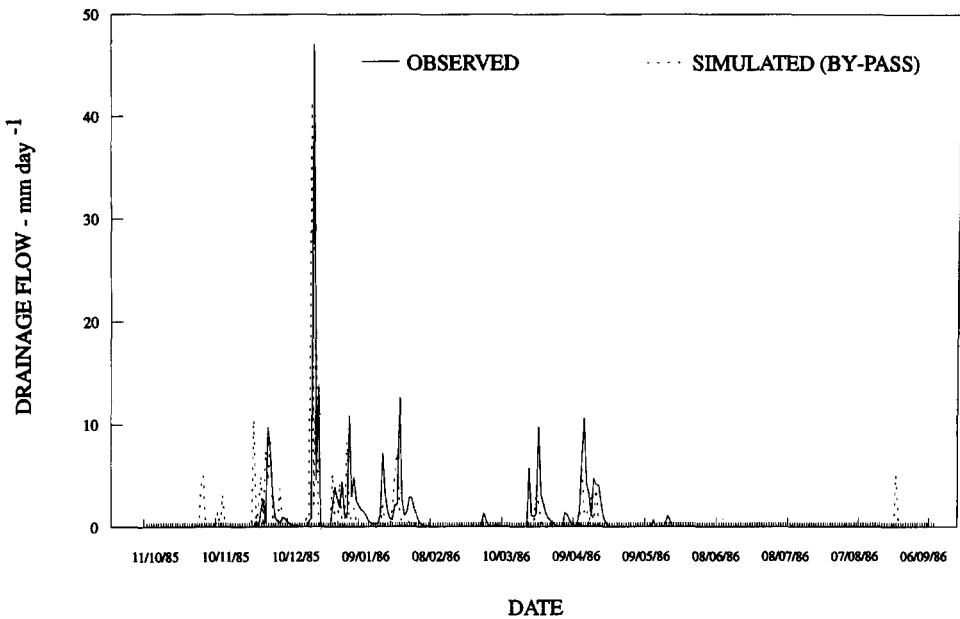


Fig. 7. Measured and simulated drainage with bypass flow – Plot 10.

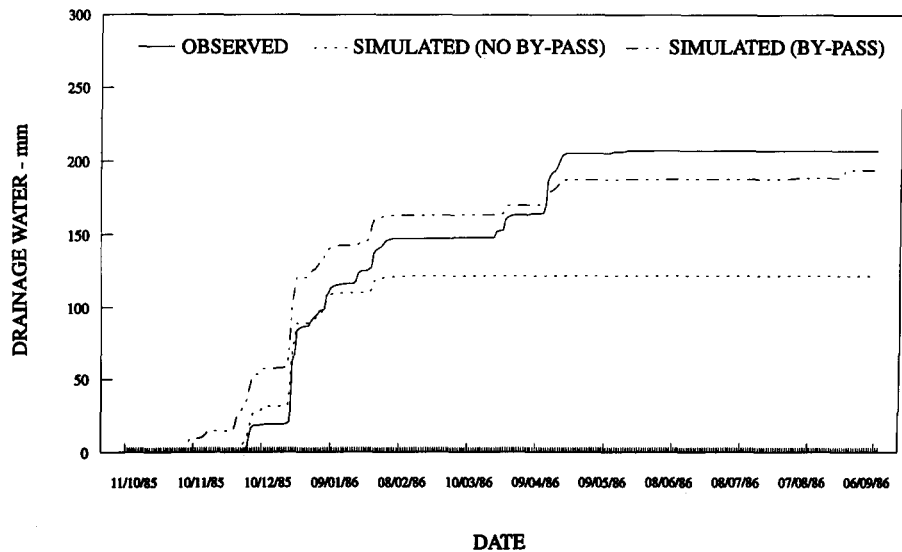


Fig. 8. Cumulative drainage – Plot 7.

Cumulative simulated drainage from Plot 7 over the year with and without bypass flow is shown in Fig. 8. The observed cumulative drainage was 207 mm whereas the simulated value was 194 mm when bypass flow was considered and 122 mm with matrix flow alone. The total soil water storage in the root zone (0–60 cm) was also simulated with and without bypass flow. Fig. 9 shows that a closer agreement between

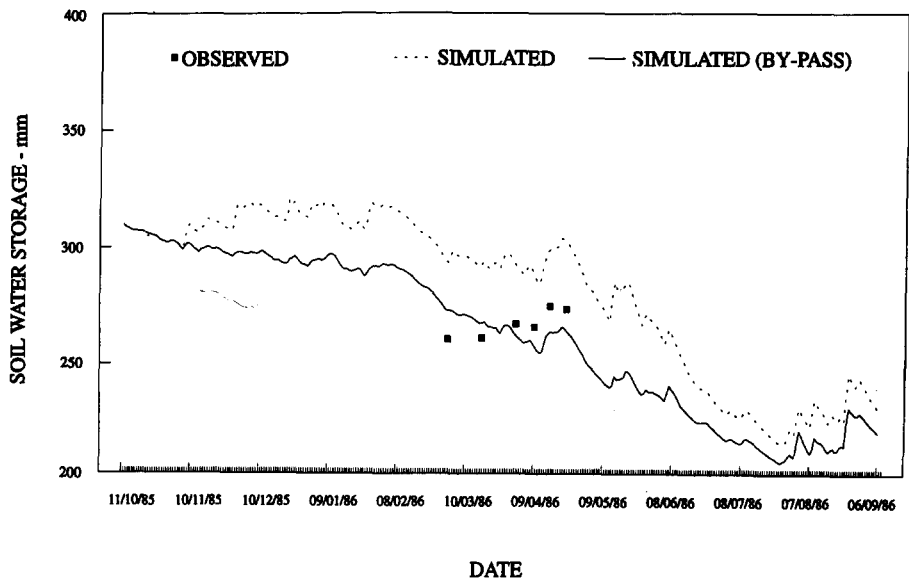


Fig. 9. Measured and simulated total soil water (0–60 cm) – Plot 7.

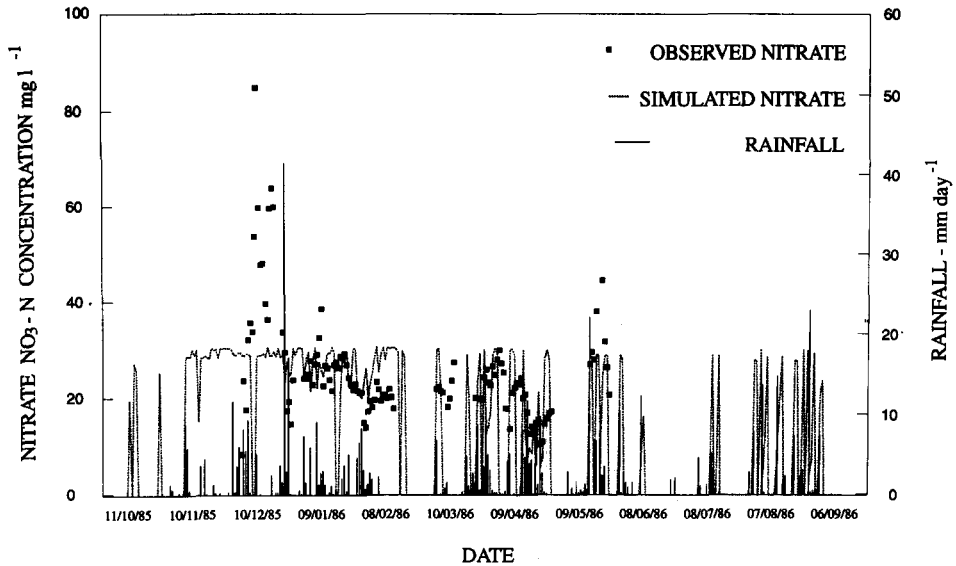


Fig. 10. Rainfall, measured and simulated nitrate ($\text{NO}_3\text{-N}$) – Plot 7.

the observed and the simulated total water storage was achieved when bypass flow was considered.

These results indicate the presence of macropore flow in the soil at Brimstone Farm and show that the SOIL model can predict both drain flow and total soil moisture storage in the root zone.

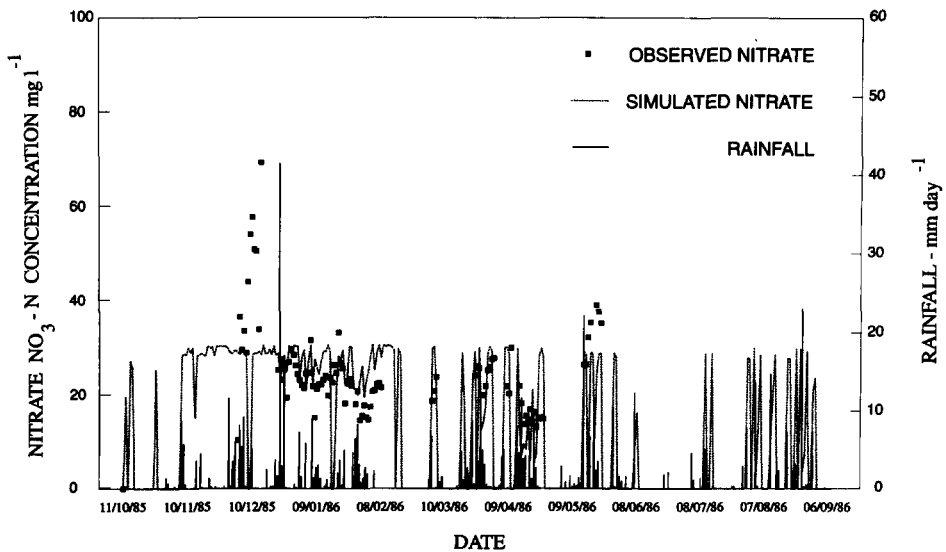


Fig. 11. Rainfall, measured and simulated nitrate ($\text{NO}_3\text{-N}$) – Plot 10.

Regularly monitored water table depths during the growing season were not available for the present modelling study. The simulated water table nearest to the surface was at 40 cm, midway between the drains. Harris et al. (1984) monitored the water table level in previous years and the value nearest to the surface that they obtained was also 40 cm.

5.2. Nitrate leaching

At this stage, the output from the SOIL model run including bypass flow was considered adequate for use as input in the SOILN model.

Rainfall and comparisons of simulated and observed nitrate leaching to field drains for Plots 7 and 10 are shown in Figs. 10 and 11, respectively. Figs. 12 and 13 show the same nitrate variables, with rainfall replaced by measured drainflow. Apart from two periods, the first between December 8 and 14, 1985 and the second between May 12 and 22, 1986, the simulated and observed nitrate concentrations for both plots were in reasonable agreement.

The high concentrations of nitrate measured in both plots from December 8 to 14 were associated with low rainfall (5.4 mm). The drainage flows during this period were also low, with a total of 0.96 mm drainage from Plot 7 and 4.6 mm from Plot 10. This may be compared with daily values of 7.6 and 9.72 mm for Plots 7 and 10 on 5 December generated by over 40 mm rainfall from 29 November to 6 December. During this first week, the first major drainage event of the winter, more detailed hourly measurements show that nitrate concentrations as $\text{NO}_3\text{-N}$ remained at around 7 mg l^{-1} . Because of the low drainage flows during the second week of December, the nitrate load associated with the high concentrations was not large, as shown in Fig. 8. Had the high nitrate concentrations been associated with high flows the poor model

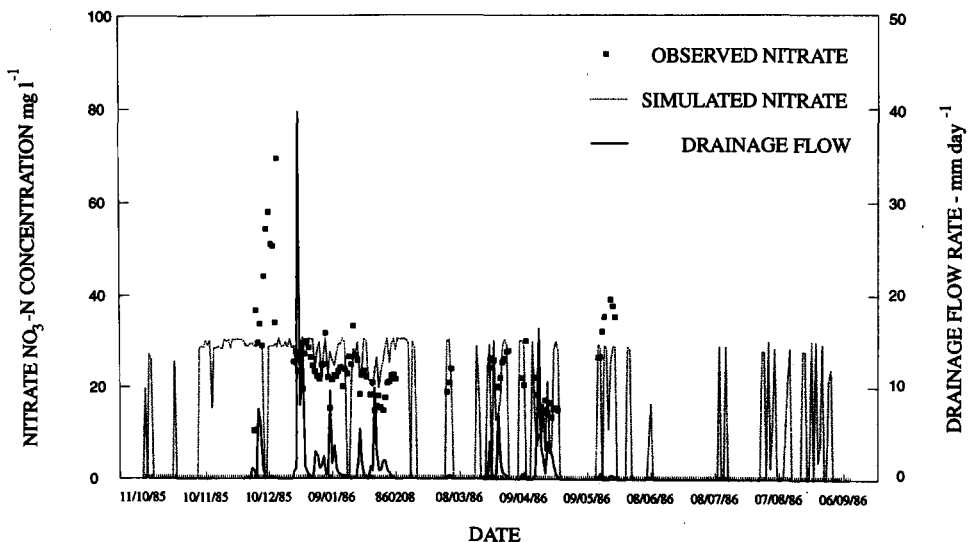


Fig. 12. Drainage flow rate, measured and simulated nitrate ($\text{NO}_3\text{-N}$) – Plot 7.

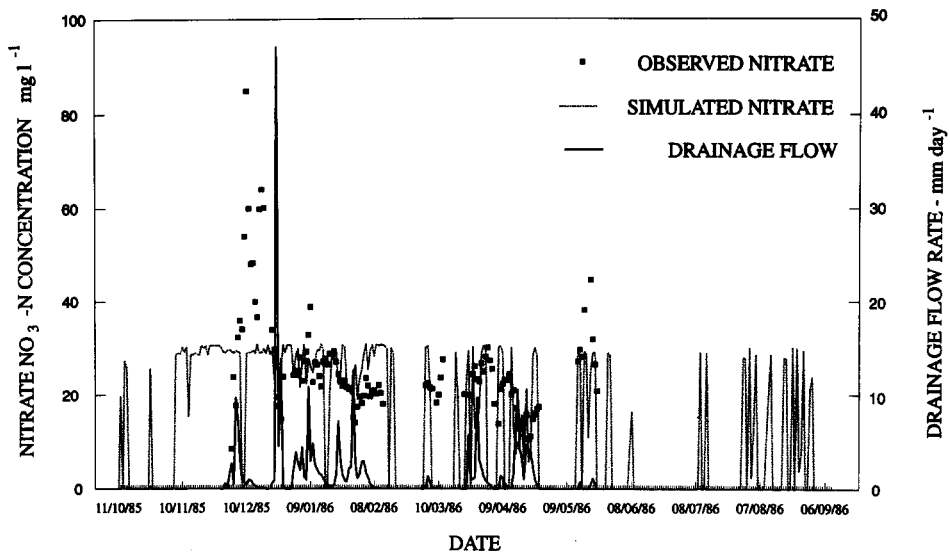


Fig. 13. Drainage flow rate, measured and simulated nitrate ($\text{NO}_3\text{-N}$) – Plot 10.

performance over this period would have been more notable. The main nitrate load is both simulated and observed during the period of high flows in late December. A possible explanation for the observed nitrate concentrations in the first 2 weeks of December is related to the influence of bypass flow. Greater drain flows during the first week are likely to have been predominantly bypass flow, not picking up much of the nitrate formed in the soil matrix by mineralisation. The smaller drainflows in the second week may have had a greater proportional contribution of matrix flow, and this picked up more of the nitrate generated by mineralisation.

High measured nitrate concentrations between May 12 and 22 may be due to leaching of fertiliser applied 2 weeks earlier on April 30. This is also a period with low drainage flows; 1.8 mm and 2.74 mm from Plots 7 and 10, respectively. These flows were generated by 45.4 mm rainfall. Again, because of the associated low flows, the high nitrate concentrations contribute little to the annual nitrate load in drainage.

It is likely that some improvement in the simulation of nitrate concentrations could be achieved by calibration of rate parameters, but our preference is to present results obtained using accepted values derived from measurements and experimental work elsewhere.

The cumulative simulated mineralisation of the litter and humus was 8.57 g N m^{-2} , with 7.01 g N m^{-2} simulated cumulative nitrification, neither of which can be tested against measurements. The total simulated N uptake by the crop including roots over the growing season amounted to 28.4 g N m^{-2} (284 kg N ha^{-1}). The field estimate of N uptake was 180 kg N ha^{-1} in grain and 40 kg N ha^{-1} in straw. The sum of these is 220 kg N ha^{-1} , should be close to the simulated value if root N was estimated.

The total nitrate leached from Plots 7 and 10 during the period of drainflow were $43.31 \text{ kg NO}_3\text{-N ha}^{-1}$ and $56.57 \text{ kg NO}_3\text{-N ha}^{-1}$, respectively, whereas the simulated loss was $44.17 \text{ kg-NO}_3\text{-N ha}^{-1}$ (Fig. 14). The simulated values were 1.99% higher

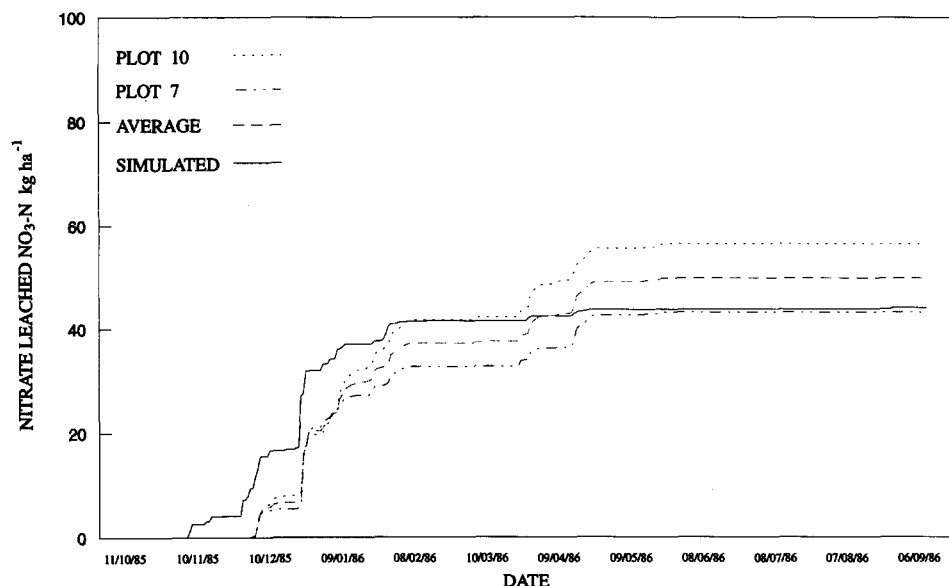


Fig. 14. Cumulative leached nitrate ($\text{NO}_3\text{-N}$) – Simulated, Plot 7, Plot 10 and average of Plot 7 and Plot 10.

than the observed for Plot 7 but 21.82% less than that for Plot 10. The mean difference between simulated and observed for both plots was therefore only -9.97% .

6. Conclusion

The results of this investigation have provided a quantitative description, through the use of S_{mat} , of the relative contribution of both matrix and bypass flow in the structured clay soil of Brimstone Farm. Bypass flow contributes to the quick response of the drainage hydrographs. The soil water balance components such as drain flow and soil water storage in the root zone were better simulated by the SOIL model when bypass flow was considered. Simulated nitrate leached to the drains using SOILN was in general well simulated subject to the chosen initial conditions, but there was a mismatch between simulated and measured data during periods of high nitrate concentration with small volumes of drainage water. The total simulated nitrogen uptake by the crop was close to the observed value.

The exercise of applying the SOIL and SOILN models to these historic data not collected specifically for modelling has also demonstrated the need for field data to be collected with close attention to the needs of the modeller. Perhaps the most important of these is a need for measurements of internal variables whose values are simulated by the model. These would give a test of model structure which would not be provided by comparing simulations and measurements of output variables (such as drain flows and concentrations) alone. In the particular case of the study considered here, this would include measurements of soil water content and nitrate concentration within each plot.

One of the features of the Brimstone data which has perhaps not been fully exploited in this or other studies is its replicated data, which in principle gives the possibility of a statistical description of model performance. A merging of the designed experimental approach common to agricultural trials with modelling in a more or less mechanistic framework is likely to greatly strengthen our understanding of the stochastic nature of subsurface hydrological and hydrochemical processes under varied agricultural conditions.

Acknowledgements

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References

- Brooks, R.H. and Corey, A.T., 1964. Hydraulic properties of porous media. Colorado State University, Colorado, Hydrology Paper No. 3, 27 pp.
- Castle, D.L., 1991. This is Brimstone. In: J.A. Catt (Editor), *The Brimstone Experiment*, Lawes Agricultural Trust, Harpenden, pp. 1–14.
- Catt, J., (Editor), 1991. *The Brimstone Experiment*. Lawes Agricultural Trust, Harpenden, 85 pp.
- Clapp, R. and Hornberger, G.M., 1978. Empirical equations for some soil hydraulic properties. *Water Resour. Res.*, 14: 601–604.
- Goss, M.J., Harris, G.L. and Howse, K.R., 1984. Functioning of mole drains in a clay soil. *Agric. Water Manage.*, 6: 27–30.
- Goulding, K.W.T., 1990. Nitrogen deposition to land from the atmosphere. *Soil Use Manage.*, 6: 61–63.
- Haria, A.H., Johnson, A.C., Bell, J.P. and Batchelor, C.H., 1994. Water movement and isoproturon behaviour in a drained heavy clay soil: 1. Preferential flow processes. *J. Hydrol.*, 163: 203–216.
- Harris, G.L. and Howse, K.R., 1991. Hydrology and soil physics studies. In: J. Catt (Editor), *The Brimstone Experiment*. Lawes Agricultural Trust, UK, pp. 15–25.
- Harris, G.L., Goss, M.J., Dowdell, R.J., Howse, K.R. and Morgan, P., 1984. A study of mole drainage with simplified cultivation for autumn sown crops on a clay soil. 2. Soil water regimes, water balances and nutrient loss in drain water, 1978–80. *J. Agric. Sci.*, 102: 561–581.
- Harris, G.L., Howse, K.R. and Pepper, T.J., 1993. Effects of moling and cultivation on soil-water and runoff from a drained clay soil. *Agric. Water Manage.*, 23: 161–180.
- Jansson, P.-E., 1991. Simulation model for soil water and heat conditions. Description of SOIL model. Swedish University of Agricultural Sciences, Department of Soil Sciences, Uppsala, Sweden, Report No. 165, 73 pp.
- Johnsson, H., Bergstrom, L., Jansson, P.-E. and Paustian, K., 1987. Simulated nitrogen dynamics and losses in a layered agricultural soil. *Agric. Ecosyst. Environ.*, 18: 333–356.
- Monteith, J.L., 1965. Evaporation and environment. In: G.E. Fogg (Editor), *The State of Movement of Water in Living Organisms*. Soc. Exp. Biol. Symp. No. 19, Cambridge University Press, London, pp. 205–234.
- Youngs, E.G. and Goss, M.J., 1988. Hydraulic conductivity profiles of two clay soils. *J. Soil Sci.*, 39: 341–345.